Q3. [26 pts] Short Answers

(a) [5 pts] Given that (x_1, x_2) are jointly normally distributed with $\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$ and $\Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{21} & \sigma_2^2 \end{bmatrix}$ ($\sigma_{21} = \sigma_{12}$), give an expression for the mean of the conditional distribution $p(x_1|x_2 = a)$.

This can be solved by writing $p(x_1|x_2 = a) = \frac{p(x_1, x_2=a)}{p(x_2=a)}$. x_2 being a component of a multivariate Gaussian is a univariate Gaussian with $x_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$. Write out the Gaussian densities and simplify (complete squares) to see the following:

$$x_1|x_2 = a \sim \mathcal{N}(\bar{\mu}, \bar{\sigma}^2), \quad \bar{\mu} = \mu_1 + \frac{\sigma_{12}}{\sigma_{22}^2}(a - \mu_2)$$

(b) [4 pts] The logistic function is given by $\sigma(x) = \frac{1}{1+e^{-x}}$. Show that $\sigma'(x) = \sigma(x)(1 - \sigma(x))$.

$$\sigma'(x) = \frac{e^{-x}}{(1+e^{-x})^2} = \frac{1}{(1+e^{-x})} \cdot \frac{e^{-x}}{(1+e^{-x})} = \left(\frac{1}{1+e^{-x}}\right)\left(1 - \frac{1}{1+e^{-x}}\right) = \sigma(x)(1 - \sigma(x))$$

(c) Let X have a uniform distribution

$$p(x; \theta) = \begin{cases} \frac{1}{\theta} & 0 \leq x \leq \theta \\ 0 & \text{otherwise} \end{cases}$$

Suppose that n samples $x_1, \ldots, x_n$ are drawn independently according to $p(x; \theta)$.

(i) [5 pts] The maximum likelihood estimate of θ is $x_{(n)} = \max(x_1, x_2, \ldots, x_n)$. Show that this estimate of θ is biased.

Biased estimator: $\hat{\theta}$ (the sample estimate) is a biased estimator of θ (the population distribution parameter) if $E[\hat{\theta}] \neq \theta$.

Here $\hat{\theta} = x_{(n)}$. And $E[x_{(n)}] = \frac{n}{n+1}\theta \neq \theta$. The steps for finding $E[x_{(n)}]$ are given in the solutions of Homework 2, problem 5(c).

(ii) [2 pts] Give an expression for an unbiased estimator of θ.

$$\hat{\theta}_{unbiased} = \frac{n+1}{n} x_{(n)}$$

$$E[\hat{\theta}_{unbiased}] = E[\frac{n+1}{n} x_{(n)}] = \frac{n+1}{n} E[x_{(n)}] = \frac{n+1}{n} \times \frac{n}{n+1}\theta = \theta$$

(d) [5 pts] Consider the problem of fitting the following function to a dataset of 100 points $\{(x_i, y_i)\}$, $i = 1 \dots 100$:

$$y = \alpha cos(x) + \beta sin(x) + \gamma$$

This problem can be solved using the least squares method with a solution of the form:

$$\begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = (X^T X)^{-1} X^T Y$$

What are X and Y?

$$X = \begin{bmatrix} cos(x_1) & sin(x_1) & 1 \\ cos(x_2) & sin(x_2) & 1 \\ \vdots & \vdots & \vdots \\ cos(x_{100}) & sin(x_{100}) & 1 \end{bmatrix} \qquad Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_{100} \end{bmatrix}$$

(e) [5 pts] Consider the problem of binary classification using the Naive Bayes classifier. You are given two dimensional features (X_1, X_2) and the categorical class conditional distributions in the tables below. The entries in the tables correspond to $P(X_1 = x_1|C_i)$ and $P(X_2 = x_2|C_i)$ respectively. The two classes are *equally likely*.

$X_1 =$ \ Class	C_1	C_2
-1	0.2	0.3
0	0.4	0.6
1	0.4	0.1

$X_2 =$ \ Class	C_1	C_2
-1	0.4	0.1
0	0.5	0.3
1	0.1	0.6

Given a data point $(-1, 1)$, calculate the following posterior probabilities:

$P(C_1|X_1 = -1, X_2 = 1) =$ Using Bayes' Rule and conditional independence assumption of Naive Bayes

$$\frac{P(X_1=-1,X_2=1|C_1)P(C_1)}{P(X_1=-1,X_2=1)} = \frac{P(X_1=-1|C_1)P(X_2=1|C_1)P(C_1)}{P(X_1=-1|C_1)P(X_2=1|C_1)P(C_1)+P(X_1=-1|C_2)P(X_2=1|C_2)P(C_2)} = 0.1$$

$P(C_2|X_1 = -1, X_2 = 1) = 1 - P(C_1|X_2 = -1, X_1 = 1) = 0.9$